

# SafetyEye – AI Powered Workplace Occupancy & Safety Monitor

## Milestone 4 Report

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**Program:** Infosys Virtual Internship

**Project:** SafetyEye – AI Powered Workplace Occupancy & Safety Monitor

**Framework:** Streamlit, YOLOv8 (Ultralytics), OpenCV

**Hardware Used:** NVIDIA GTX 1650 Ti GPU (4GB)

**Software:** Python, PyTorch, Ultralytics YOLOv8, OpenCV, Streamlit, VS Code

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## Project Overview

SafetyEye is an AI-powered workplace monitoring system designed to improve safety compliance in construction and industrial environments. The system uses computer vision and deep learning techniques to automatically detect safety-related objects such as helmets, safety vests, workers, and machinery.

While earlier milestones focused on model training and basic detection scripts, Milestone 4 represents the final phase where the project was transformed into a complete, user-friendly application featuring a modern dashboard, real-time analytics, and comprehensive incident logging.

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## Milestone 4 Objectives

The main objectives of Milestone 4 were:

- Developing a professional Dashboard UI using Streamlit
  - Integrating the trained YOLOv8 model with the live UI
  - Implementing a real-time Alert System for safety violations
  - Creating a comprehensive Analytics Page for data visualization
  - Developing a Logs Management System with export capabilities
  - Modularizing the codebase for better maintenance
  - Performing complete system testing and performance optimization
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## Dashboard and UI Development

### Objective

The objective of Week 7 was to design and build the core user interface of the SafetyEye monitoring system to move away from command-line execution and provide a standard professional dashboard.

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### Dashboard Implementation

A modern, dark-themed interactive dashboard was developed using the Streamlit framework. The UI was designed with a futuristic neon-glow aesthetic to ensure a premium feel.

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### Key UI Components

- **Sidebar Navigation:** A collapsible menu allowing users to switch between Dashboard, Webcam Monitoring, Video Testing, Analytics, and Logs
  - **Glassmorphism Design:** Semi-transparent panels with neon borders to highlight important information
  - **KPI Summary Cards:** Real-time counters for Total Persons, Unsafe Workers, and Alerts Count
  - **Dynamic Feedback:** Visual indicators for Camera Status (Active/Inactive) and AI System Status
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### Alert System Improvement

The basic text-based alerts were upgraded into a visual Alert Panel:

- **Color-Coding:** Green cards indicate “All Safe”, while Red cards indicate unsafe violations
  - **Information Display:** Each alert includes violation type, source, and timestamp
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## System Integration and Advanced Features

### Objective

The objective of Week 8 was to integrate analytics and logging modules and ensure seamless communication between all system components.

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### Data Persistence and Integration

The system uses Streamlit session state to manage logs across pages without losing data. This allows real-time data flow from the monitoring module to analytics and logs.

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### System Modularization

The project was restructured into modular components:

- **app.py:** Main entry point handling navigation and monitoring
  - **pages/analytics.py:** Handles data visualization
  - **pages/logs.py:** Handles logging and filtering
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### Performance Optimization

- Reduced latency in video processing
  - Optimized frame handling
  - Efficient session state management
  - Smooth page switching with consistent FPS
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### Dashboard Features

The dashboard provides real-time monitoring and system overview:

- Real-time metrics for persons and safety compliance
  - System status indicators
  - Recent alerts feed
  - Mini analytics charts for quick insights
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## Analytics System

The analytics page transforms logs into visual insights:

- **Violations Over Time:** Line chart showing trends
  - **Violation Distribution:** Pie chart comparing types
  - **Safety Compliance Rate:** Donut chart of safe vs unsafe events
  - **AI Insights Panel:** Summary of trends and key observations
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## Logs Management System

The logs page maintains detailed records of safety events:

- **Data Fields:** Timestamp, violation type, source, status, confidence
  - **Filtering Options:** Date, type, and source
  - **Search Functionality:** Quick lookup of records
  - **Export Feature:** Download logs as CSV
  - **Visual Styling:** Color-coded rows (Red = unsafe, Green = safe)
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## System Architecture and Integration

The SafetyEye system operates as a unified pipeline:

- **Input Layer:** Captures frames from webcam or video
  - **Processing Layer:** Frames passed to YOLOv8 for detection
  - **Logic Layer:** Identifies missing safety equipment
  - **Storage Layer:** Stores results in session-state logs
  - **Presentation Layer:** Displays video, alerts, and analytics
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## Testing and Validation

The system was thoroughly tested:

- **Webcam Testing:** Verified real-time detection under different conditions
- **Video Testing:** Tested using pre-recorded dataset videos
- **UI Stress Testing:** Ensured stability during rapid page switching

- **Data Validation:** Confirmed accuracy of logs and analytics
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**Observations**

- Higher accuracy when subjects face the camera
  - Real-time charts improve situational awareness
  - Modular structure simplifies debugging and feature addition
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**Challenges Faced**

- Performance lag during real-time processing
  - Session state issues causing data reset
  - UI styling conflicts with Streamlit defaults
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**Solutions Implemented**

- Optimized frame processing and inference loop
  - Used session state for persistent data storage
  - Applied custom CSS for consistent UI design
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**Performance Summary**

| Feature                           | Status   | Performance Metric |
|-----------------------------------|----------|--------------------|
| Live Inference (GPU)              | Complete | ~25–30 FPS         |
| Inference (CPU)                   | Complete | ~5–8 FPS           |
| UI Navigation Speed               | Complete | <0.5 sec           |
| Log Export Accuracy               | Complete | 100% Match         |
| Filtering Latency                 | Complete | <0.1 sec           |
| Detection Accuracy (mAP) Verified |          | ~0.32              |

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## System Evolution Comparison

| Milestone | Features                     | Output Type       | System Level     |
|-----------|------------------------------|-------------------|------------------|
| M1        | Dataset + Initial Training   | Raw Predictions   | Basic            |
| M2        | Model Optimization           | Improved Accuracy | Intermediate     |
| M3        | Real-Time Detection          | Alerts + Webcam   | Advanced         |
| M4        | Dashboard + Analytics + Logs | Full Application  | Production-Level |

## Future Improvements

- Email/SMS alerts for critical violations
- Multi-camera monitoring support
- Database integration for persistent storage

## Milestone 4 Results

Key outcomes achieved:

- Professional dashboard UI developed
- Real-time webcam and video monitoring integrated
- Advanced analytics with interactive charts implemented
- Logs system with filtering and export functionality created
- Modular code structure achieved
- System performance optimized for smooth operation

## Milestone 4 Conclusion

Milestone 4 successfully transformed the SafetyEye project from a detection model into a complete, professional-grade workplace safety monitoring system. By integrating YOLOv8 with a modern Streamlit dashboard, the system now provides real-time monitoring, intelligent analytics, and structured incident logging.

The system demonstrates how AI can be effectively deployed in real-world environments to enhance workplace safety, reduce human error, and enable proactive risk management. The project is now complete and ready for deployment.